Lab 15 — Windows Pod Networking Deep-Dive — HNS & Calico for Windows

K8S-LAB-15 · Windows · Engineer · 90 min

Focus: HNS, Calico-Windows, VFP, Windows kube-proxy, PodCIDR, DNS

Objective

Inspect HNS networks/endpoints, install Calico for Windows, validate Pod-to-Pod and Pod-to-Service traffic across Linux and Windows nodes.

Concepts Covered

• HNS networks (l2bridge, overlay) and HNS endpoints

• VFP (Virtual Filtering Platform) — Windows equivalent of iptables/eBPF

• How kube-proxy on Windows programs VFP rules

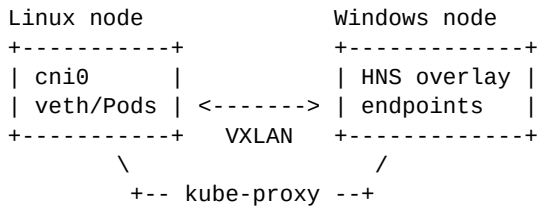
• PodCIDR allocation differences

Prerequisites

• Lab 11 (Windows worker joined)

• Cluster CIDR known

Architecture



Lab Steps

01. Install Calico for Windows

```
.\install-calico-windows.ps1 -KubeVersion 1.30.0 -ServiceCidr 10.96.0.0/12 -DNSServerIPs 10.96.0.10
```

Expected: Services CalicoNode, CalicoFelix, kube-proxy Running.

02. Inspect HNS objects

```
Get-HnsNetwork | Select Name,Type,AddressPrefix
Get-HnsEndpoint | Select Name,IPAddress
Get-HnsPolicyList
```

Expected: Overlay network present; one endpoint per Pod.

03. Validate Pod-to-Pod across OS

```
kubectl run linbox --image=nicolaka/netshoot -- sleep 3600
kubectl run winbox --image=mcr.microsoft.com/windows/nanoserver:ltsc2022 --overrides='{"spec":{"nodeSelecto
kubectl exec -it linbox -- ping <winbox-pod-ip>
```

Expected: Replies in both directions.

04. Validate Service DNS from Windows

```
kubectl expose pod linbox --port=80 --name=linbox-svc
kubectl exec -it winbox -- cmd /c "nslookup linbox-svc.default.svc.cluster.local 10.96.0.10"
```

Expected: Resolves to ClusterIP.

Common Errors

Windows Pods get IPs but can't reach Linux Pods
Cause: VXLAN UDP/4789 blocked by Windows firewall
Fix: New-NetFirewallRule -Protocol UDP -LocalPort 4789 -Action Allow

Pod stuck ContainerCreating with HNS error
Cause: Stale HNS endpoint
Fix: Remove-HnsEndpoint and recreate Pod

DNS lookups fail only from Windows
Cause: kube-proxy.exe not running or wrong DNS IP
Fix: Re-run installer with correct -DNSServerIPs

Troubleshooting

• Get-HnsNetwork | ConvertTo-Json -Depth 10

• vfpctrl.exe /list-vmswitch-port

• crictl ... logs <pod>

Enterprise Relevance

HNS/VFP knowledge is the difference between a Windows node that 'mostly works' and one that survives upgrades and reboots.

Validation

• Linux Pod <-> Windows Pod ping works both ways

• Windows Pod resolves Linux Service via cluster DNS

• Calico services healthy after node reboot

Questions

[Beginner] Q: Linux equivalent of an HNS endpoint?

A: A veth pair plugged into the CNI bridge.

[Intermediate] Q: What does VFP do?

A: Programs L3/L4 filtering, NAT, LB rules inside the Hyper-V virtual switch.

[Advanced] Q: Why is Restart-Service hns dangerous?

A: Tears down every Pod's networking on the host.

[Scenario] Q: After reboot, all Pods stay ContainerCreating. First check?

A: Get-HnsNetwork — overlay missing means Calico service didn't reattach.

[Viva] Q: How does kube-proxy work on Windows?

A: kube-proxy.exe programs VFP DNAT/LB rules instead of iptables.